

Ayush Maheshwari

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Experience

NVIDIA

Senior Solutions Architect, NVAITC

Gurugram

Oct 2024 – Present

- Foundation Models: Building multilingual and domain-specific foundation models for Indian languages and scientific domains.
- Research Collaborations: Collaborating with premier academic institutions and research organizations.

Vizzhy Inc

Lead Research Scientist, Language Model Team

Bengaluru

Sep 2023 – Sep 2024

- Led a team of 5 people to build Indic large language models from scratch.
- Training the model on large accelerator cluster.

Adobe Research

Research Intern, Document Cloud Team

Bengaluru

May – Aug 2021

NCETIS, IIT Bombay

Project Engineer

Mumbai

May 2016 – Dec 2018

Water & Climate Lab, IIT Gandhinagar

Junior Research Fellow

Gandhinagar

May 2015 – Apr 2016

Tata Consultancy Services

Systems Engineer

Mumbai

Oct 2011 – June 2013

Education

Indian Institute of Technology Bombay

Ph.D. in Computer Science and Engineering

Mumbai, India

Jan 2019 – Aug 2023

Defence: July 2024; 'Impactful Research Award 2023' by IIT Bombay

Thesis: Knowledge Integration in NLP Models using Constraint Ingestion and Generation

Advisor: Prof. Ganesh Ramakrishnan

Tata Institute of Social Sciences

M.Sc.

Mumbai, India

2013 – 2015

University College of Engineering

B.Tech. (CSE)

Rajasthan Technical University Kota, India

2007 – 2011

Featured Publications

- LexGen: Domain-aware Multilingual Lexicon Generation, ACL (Main), 2025
Maheshwari, A., Singh A., Karthika NJ, Bhat K., Jyothi P., Ramakrishnan G. 

2. DictDis: Dictionary Constrained Disambiguation for Improved NMT, Findings of EMNLP, 2024 **Maheshwari, A.**, Jyothi P., Ramakrishnan G. [↗](#)
3. Adaptive mixing of auxiliary losses in supervised learning, AAAI, 2023 ([Oral Presentation](#)) [↗](#)
Sivasubramanian D., **Maheshwari, A.**, Prathosh AP, Shenoy P., Ramakrishnan G.
4. Learning to Robustly Aggregate Labeling Functions for Semi-supervised Data Programming, Findings of ACL, 2022 [↗](#)
Maheshwari, A., Killamsetty K., Ramakrishnan G., Iyer R., Danilevsky M., Popa L.
5. Data Programming using Semi-Supervision and Subset Selection, Findings of ACL, 2021 [↗](#)
Maheshwari, A., Chatterjee O., Killamsetty K., Iyer R., Ramakrishnan G.
6. Joint Learning of Hyperbolic Label Embeddings for Hierarchical Multi-label Classification, EACL 2021 [↗](#)
Maheshwari A., Chatterjee S., Ramakrishnan, G. Jagaralpudi S.
7. UDAAN-Machine Learning based Post-Editing tool for Document Translation, CODS-COMAD Demonstrations, 2023 ([Best Demo Paper Award](#)) [↗](#)
Maheshwari, A., R. Ajay., Subramanian V., Ramakrishnan G.

Open-Source contributions

- **SPEAR** : Data-Programming python library to create training data. [GitHub](#)
- **UDAAN** : Post-editing workbench to quickly create publishable translations. [Website](#)

Research Interests

- Develop large language models for mid and low-resource languages.
- Develop machine learning based methods that generalize to new domains with limited resources.
- Lexically constrained neural machine translation, data programming, semi-supervised learning

Awards, Services & Talks

- My PhD thesis work was awarded **Impactful Research Award 2023** by IIT Bombay
- Best Demo Paper Award at CODS-COMAD 2023
- Awarded prestigious Ekal Fellowship 2020
- Reviewer: ARR 22–24, EACL, EMNLP, ACL (21–23)
- Fully covered grant for attending EMNLP 2022, Abu Dhabi by EMNLP DEI Committee
- Co-Founder at non-profit [Temples of India](#) - A web and app platform listing > 700K temples across Indian subcontinent.
- Key Member of [Udaan Project](#) - A project to translate education textbooks in Indian languages.

Computer Skills

Programming Languages: Python, C, C++, Javascript/jQuery, Bash, SQL

Libraries: PyTorch, DeepSpeed, HuggingFace, NumPy, scikit-learn

Talks

- Delivered Masterclass on **Building data-efficient large language models**
[NLTF NASSCOM](#), 2024
- Invited Tutorial on **Data Efficient Machine Learning for end-to-end Educational Content Creation**
[International Conference on Educational Data Mining](#), 2023
- Talk on **SPEAR: Data Programming using Semi-Supervision** [\[Talk\]](#)
[DECILE Workshop](#), 2022